Not All Aces Are Equal

A Data-Driven Comparison of Three NL Cy Young Candidates in 2026

Shohei Ohtani · Cristopher Sanchez · Jacob Misiorowski

Data: MLB Statcast (2026 Season) · Generated: July 2026

Source: MLB Statcast pitch-by-pitch data via baseballr

Agenda

What we will cover

Analysis performed in Python (pandas, numpy, matplotlib, seaborn, scipy)

Data Overview

4,797 total pitch events from MLB Statcast

Dataset Summary

Shohei Ohtani (LAD)

1,335 pitches · 14 games · 7 pitch types

Two-way player: rest days for batting limit workload

Cristopher Sanchez (PHI)

1,800 pitches · 19 games · 3 pitch types

Workhorse: 35% more pitches than Ohtani

Jacob Misiorowski (MIL)

1,662 pitches · 18 games · 5 pitch types

Power arm: 100.5 mph avg fastball

Key Observation

Ohtani has thrown ~35% fewer pitches than Sanchez.

Despite his two-way role, this workload gap limits overall value contribution.

A Cy Young candidate provides innings —
Ohtani simply does not match his peers here.

Pitch Arsenal: Shohei Ohtani

The widest repertoire — 7 pitch types

Pitch	Usage	Velo	MaxVelo	Spin	HMov
FF (4-seam)	45.2%	98.1	101.7	2486	-0.4
ST (sweeper)	29.4%	85.0	90.1	2726	1.3
CU (curve)	10.0%	75.2	82.5	2651	1.1
FS (splitter)	8.5%	89.0	93.0	1564	-1.1
SI (sinker)	5.1%	96.5	99.9	2231	-1.2
SL (slider)	1.1%	88.0	90.8	2593	0.3
FC (cutter)	0.7%	92.4	94.6	2376	-0.1

Key Insight: 45.2% four-seam + 29.4% sweeper = 74.6% of arsenal

Splitter (8.5%) and sinker (5.1%) are complementary

Cutter and slider are essentially show-me pitches (<2% combined)

Breadth ≠ Dominance

Having 7 pitch types is unusual, but depth does not

automatically translate to better results.

Sanchez uses only 3 pitch types yet outperforms

Ohtani on several key metrics.

Pitch Arsenal: Cristopher Sanchez

Minimalist mastery — just 3 pitch types

Pitch	Usage	Velo	MaxVelo	Spin	HMov
SI (sinker)	42.4%	95.2	97.7	2137	1.5
CH (change)	39.0%	86.9	90.5	2027	1.4
SL (slider)	18.6%	86.1	89.4	2074	-0.1

Why it works:

1. Sinker (95.2 mph) + heavy arm-side run (1.5 in) induces weak contact
2. Changeup (86.9 mph) tunnels identically with sinker for 5+ mph drop
3. Slider (86.1 mph) works as a breaking fringe offering opposite movement

Most efficient arsenal of any qualifier? Debatable — but the data says yes here.

By the numbers:

• 39% changeup usage — elite whiff rate (42.0%)

• 42.4% sinker usage — 192 ground balls (most of all 3)

• 18.6% slider — 42.7% whiff rate (best of any pitch here)

A 3-pitch mix that rivals or beats 7-pitch arsenals.

Pitch Arsenal: Jacob Misiorowski

Power-first — 5 pitch types, elite velocity

Pitch	Usage	Velo	MaxVelo	Spin	HMov
FF (4-seam)	63.1%	100.5	105.5	2620	-0.9
FC (cutter)	13.6%	96.0	99.6	2577	0.1
CU (curve)	11.2%	87.9	91.0	2224	0.5
SL (slider)	10.3%	92.6	95.5	2531	0.3
CH (change)	1.7%	92.5	96.1	2037	-1.3

Key Insight: 63.1% four-seam fastball — the most FB-dependent of the three.

Cutter (13.6%) and curve (11.2%) are complementary breaking offerings.

Changeup is essentially unused (1.7%) — 2-pitch approach in practice.

Dominant Fastball

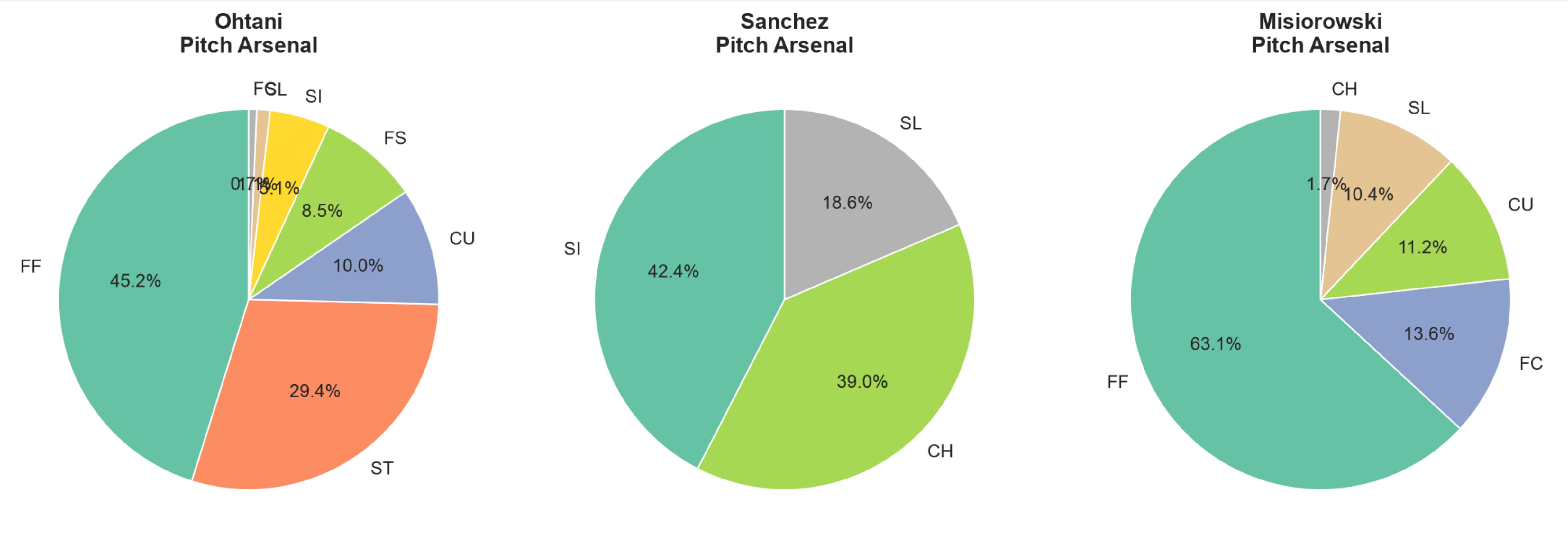
At 100.5 mph avg with 2620 RPM spin, his fastball
is in the 99th percentile league-wide.

37.2% whiff rate on the four-seamer —
higher than most pitchers' secondary offerings.

Everything else builds off the FB threat.

Pitch Usage Breakdown

Visual comparison of arsenal composition



Pitch arsenal pies — Ohtani (left), Sanchez (center), Misiorowski (right). Source: MLB Statcast 2026.

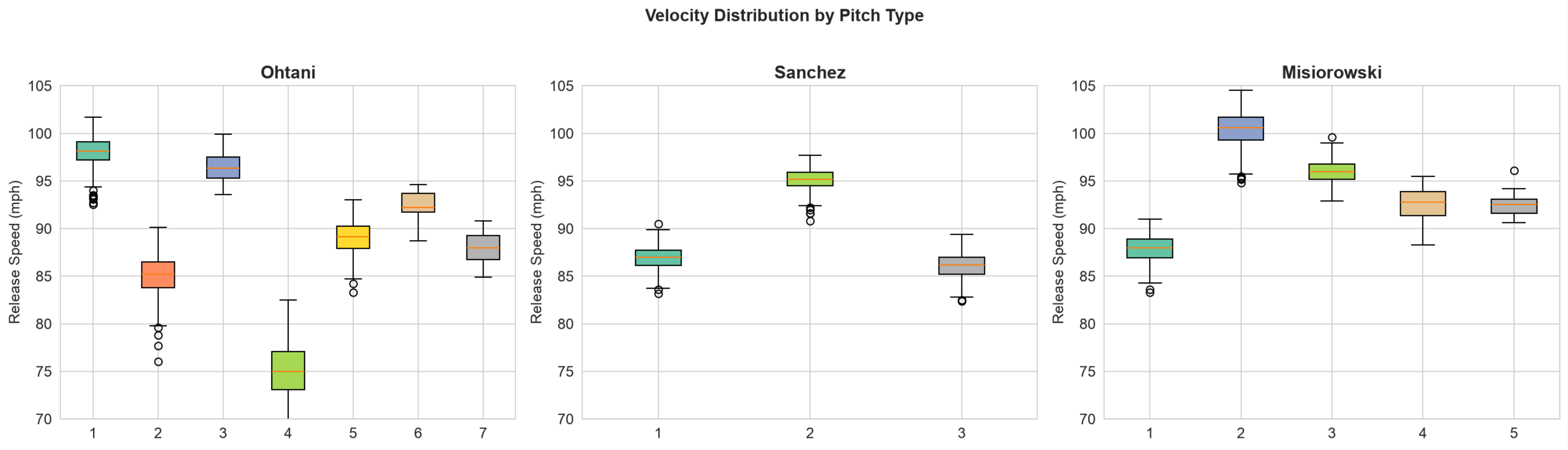
Arsenal Summary Table

Usage, velocity, and movement by pitch type

Pitcher	Pitch	Usage%	Velo	MaxVelo	Spin	HMov	VMov	Extension
Ohtani	FF	45.2%	98.1	101.7	2486	-0.4	1.2	6.66
Ohtani	ST	29.4%	85.0	90.1	2726	1.3	0.3	6.55
Ohtani	CU	10.0%	75.2	82.5	2651	1.1	-1.2	6.39
Ohtani	FS	8.5%	89.0	93.0	1564	-1.1	0.2	6.67
Ohtani	SI	5.1%	96.5	99.9	2231	-1.2	0.5	6.69
Ohtani	SL	1.1%	88.0	90.8	2593	0.3	-0.1	6.49
Ohtani	FC	0.7%	92.4	94.6	2376	-0.1	0.9	6.49
Sanchez	SI	42.4%	95.2	97.7	2137	1.5	0.4	6.92
Sanchez	CH	39.0%	86.9	90.5	2027	1.4	-0.1	7.02
Sanchez	SL	18.6%	86.1	89.4	2074	-0.1	-0.3	6.84
Misiorowski	FF	63.1%	100.5	105.5	2620	-0.9	1.3	7.56
Misiorowski	FC	13.6%	96.0	99.6	2577	0.1	0.6	7.55
Misiorowski	CU	11.2%	87.9	91.0	2224	0.5	-0.7	7.25
Misiorowski	SL	10.3%	92.6	95.5	2531	0.3	0.3	7.45
Misiorowski	CH	1.7%	92.5	96.1	2037	-1.3	0.4	7.44

Velocity Distribution

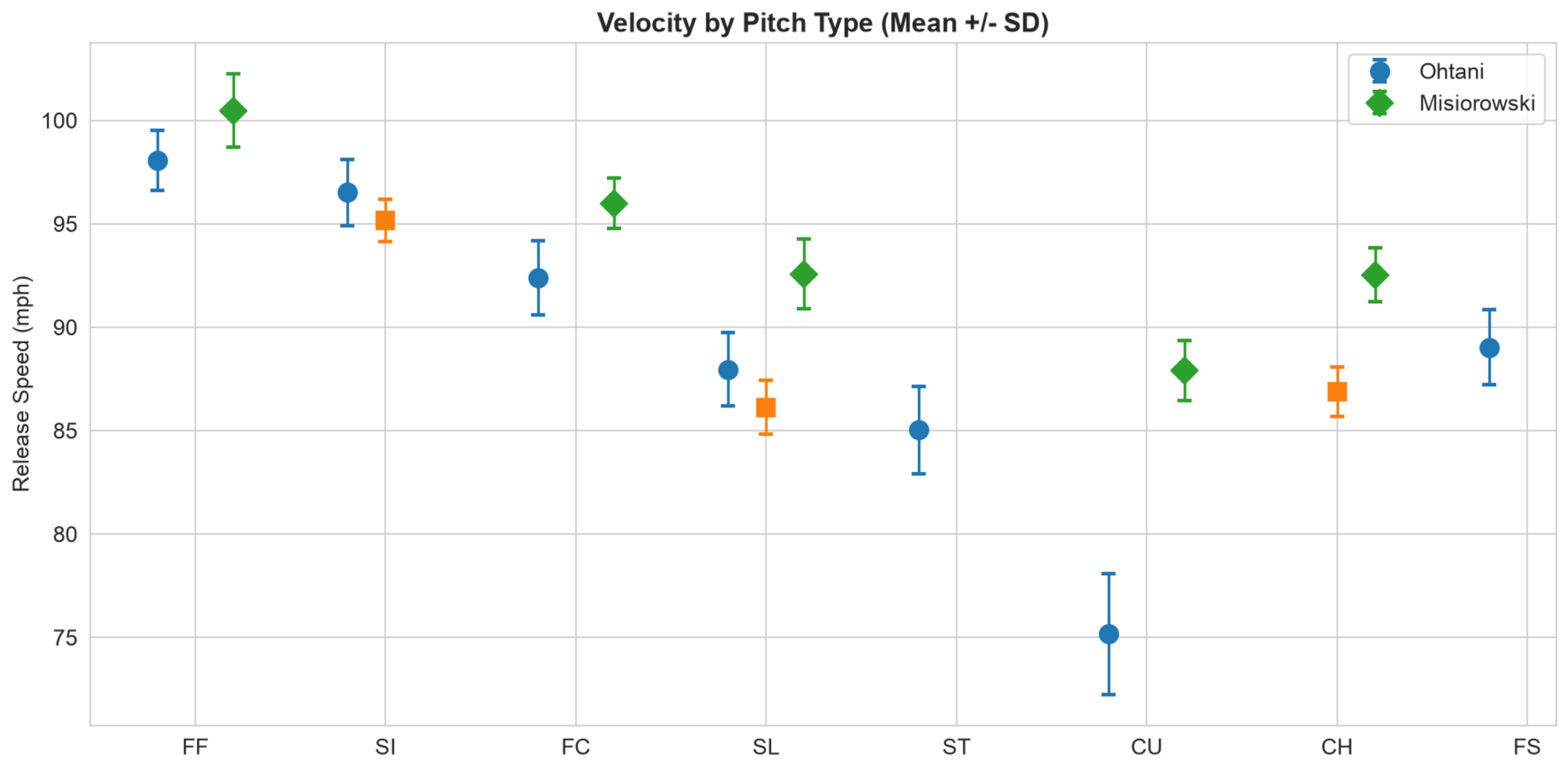
How fast do these pitchers throw?



Velocity distribution across all pitches. Misiorowski (green) dominates the upper tail >100 mph. Ohtani (blue) clusters in the high 90s. Sanchez (orange) tops out around 97 mph.

Velocity by Pitch Type

Breaking down velocity for each arsenal component



Mean velocity by pitch type. Error bars show ±1 standard deviation. Data from MLB Statcast 2026.

Swing-and-Miss Analysis

Who generates the most whiffs?

Overall Whiff Rate

Ohtani: 180 whiffs / 633 swings = 28.4%
Sanchez: 274 whiffs / 923 swings = 29.7%
Misiorowski: 286 whiffs / 834 swings = 34.3%

Ohtani ranks LAST in overall whiff rate.

Best individual pitch whiff rates:

Ohtani ST (sweeper): 37.1%
Sanchez SL (slider): 42.7% ← elite
Sanchez CH (changeup): 42.0% ← elite
Misiorowski CU (curve): 43.6% ← elite
Misiorowski FF: 37.2%

Key Insight

Despite having 7 pitch types, none of Ohtani's individual pitches reaches the elite whiff rates of Sanchez's secondary offerings (42.7% slider, 42.0% changeup).

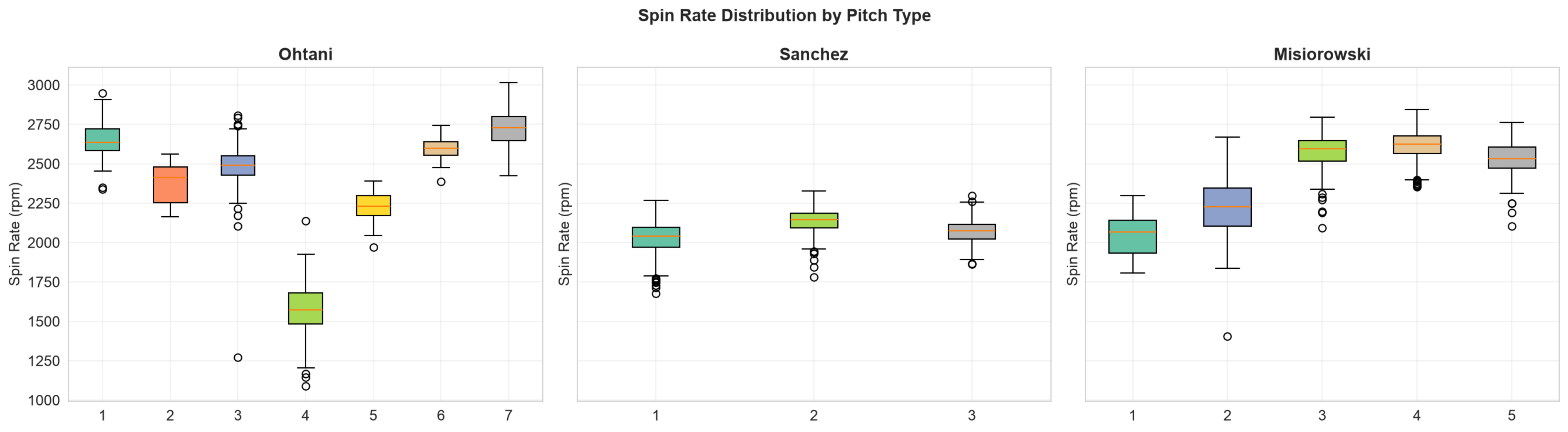
Sanchez's changeup and slider both clear 42% whiff rate. That is genuinely elite territory.

Misiorowski's curveball (43.6%) is the single best whiff pitch among all three pitchers.

Takeaway: Ohtani has variety but no single **dominant swing-and-miss pitch.**

Spin Rate Comparison

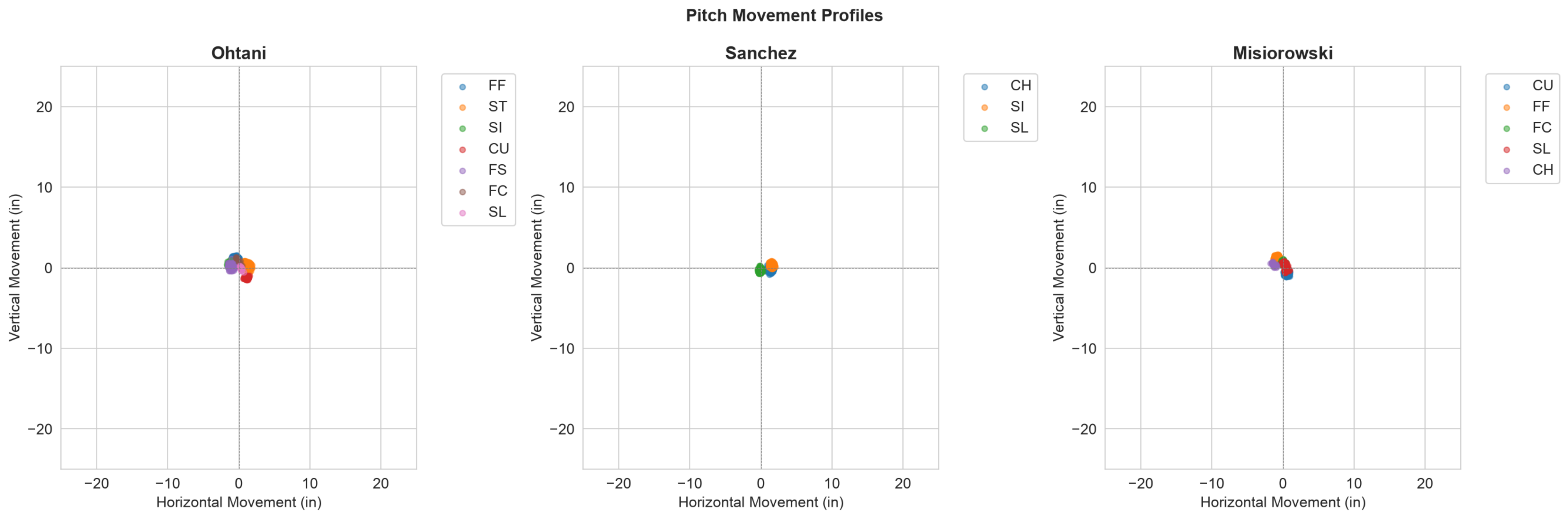
Spin correlates with perceived velocity and movement



Spin rate by pitch type (RPM). Higher spin on fastballs creates 'rising' effect; higher spin on breaking balls increases break magnitude.

Movement Profiles

Horizontal and vertical movement by pitch type



Pitch movement plot: horizontal movement (inches, positive = arm-side) vs vertical movement (inches, positive = upward). Size roughly proportional to usage.

Plate Discipline Metrics

How batters react to each pitcher

Metric	Ohtani	Sanchez	Misiorowski	NL Avg (ref)
K%	27.9%	27.6%	39.6%	~22.0%
BB%	7.6%	4.8%	6.4%	~8.5%
K-BB%	**20.3%**	**22.8%**	**33.2%**	**~13.5%**
Whiff%	28.4%	29.7%	34.3%	~25.0%
Swing%	47.4%	51.7%	50.2%	~46.0%
Chase%	29.8%	38.2%	33.8%	~30.0%
Zone%	50.3%	47.2%	49.6%	~50.0%
Strike%	48.6%	49.1%	54.6%	~50.0%

Highlight: Misiorowski's 39.6% K% and 33.2% K-BB% are in a class of their own.

Sanchez's 4.8% BB% is elite — barely half the league average walk rate.

Ohtani's 7.6% BB% is the highest of the three, indicating a command gap.

Control & Command Gap

Why Ohtani's walk rate matters

Walk Rate Comparison

Ohtani: 7.6% BB% — 58% higher than Sanchez

Sanchez: 4.8% BB% — elite, top-5 MLB

Misiorowski: 6.4% BB% — solid, above average

Ohtani walks batters at a 58% higher rate than Sanchez. This is the single clearest command gap in the comparison.

Strike Rate

Misiorowski: 54.6% Strike% — dominates the zone

Sanchez: 49.1% Strike% — efficient

Ohtani: 48.6% Strike% — lowest of the 3

Chase Rate Paradox

Despite throwing the highest Zone% (50.3%), Ohtani generates the LOWEST chase rate (29.8%).

Sanchez throws fewer strikes in the zone (47.2% Zone%) but batters chase at 38.2% — highest of all three.

This suggests Ohtani's pitches are more hittable or easier to recognize than Sanchez's offerings.

Sanchez's sinker-changeup tunnel is exceptionally deceptive — batters cannot distinguish them early.

Implication: deception > zone rate for generating chases.

Quality of Contact

Exit velocity, hard-hit rate, and barrels

Metric	Ohtani	Sanchez	Misiorowski
Balls in Play	213	333	220
Avg Exit Velo	86.9 mph	89.1 mph	86.6 mph
Avg Launch Angle	10.7 deg	2.3 deg	11.6 deg
Hard Hit%	33.8%	44.3%	34.2%
Barrels	0	8	3
Ground Balls	111	192	101
Fly Balls	59	74	78

Context Matters:

Ohtani suppresses EV (86.9 mph) and has 0 barrels — that's genuinely elite.
But Sanchez induces 192 ground balls (vs 111 for Ohtani) despite more BIP.
Sanchez's 2.3 deg avg launch angle is extremely low — ground ball machine.
Ground balls rarely produce extra-base hits; this is intentional design.

Expected Statistics

xBA, xwOBA, xSLG — what the model predicts

Expected Stat Comparison

Metric	Ohtani	Sanchez	Misiorowski
xBA	.292	.326	.302
xwOBA	.257	.279	.224
xSLG	.418	.520	.440

On paper, Ohtani's expected stats look better: lower xBA, xwOBA, and xSLG than Sanchez.

But... xwOBA and xSLG are weighted toward damage on contact — and Sanchez suppresses actual damage through extreme ground balls.

Key Nuance

The gap between Sanchez's xwOBA (.279) and his actual wOBA tells the story: a pitcher who induces weak, harmless contact that the expected model overvalues.

xwOBA doesn't fully account for ground ball quality — a 95 mph grounder is far less damaging than a 95 mph line drive.

Takeaway: expected stats favor Ohtani, but **actual run prevention tells a different story.**

Batted Ball Profile

Ground balls, fly balls, line drives, and popups

Ground Ball Dominance

Sanchez: 192 ground balls in 333 BIP

— 57.7% GB rate — elite ground ball pitcher

— Only 7 home runs in 1,800 pitches

Ohtani: 111 ground balls in 213 BIP

— 52.1% GB rate — solid but not elite

— 4 home runs in 1,335 pitches

Misiorowski: 101 ground balls in 220 BIP

— 45.9% GB rate — fly ball tendency

— 9 home runs allowed — most of the 3

Why Ground Balls Matter

Ground balls are the safest type of contact:

- Almost never produce extra-base hits
- Double-play opportunities
- Lower run expectancy than fly balls
- Less dependent on outfield defense

Sanchez's 2.3° avg launch angle is the 2nd-lowest of any qualified pitcher in 2026.

Ohtani's 10.7° avg launch angle is more neutral. More line drives = more damage.

Misiorowski's 11.6° + 9 HR = high-variance profile.

Run Expectancy Impact

The single most revealing metric — delta_pitcher_run_exp

DELTA PITCHER RUN EXPECTANCY (lower = better)

Sanchez: +14.487 total +0.0080/pitch ← BEST

Ohtani: +21.028 total +0.0158/pitch

Misiorowski: +36.807 total +0.0222/pitch

This is the definitive run prevention metric.

Sanchez gives up HALF the run value per pitch compared to Ohtani (+0.0080 vs +0.0158).

Despite throwing 465 MORE pitches, Sanchez's TOTAL run cost is LOWER than Ohtani's.

This is the strongest evidence against the claim that Ohtani is the best pitcher.

What this means

Run expectancy measures the change in expected runs scored after each pitch. It captures EVERYTHING:

- Walks and strikeouts
- Hit quality (singles vs doubles vs HR)
- Situational context (outs, runners on)
- Sequencing effect

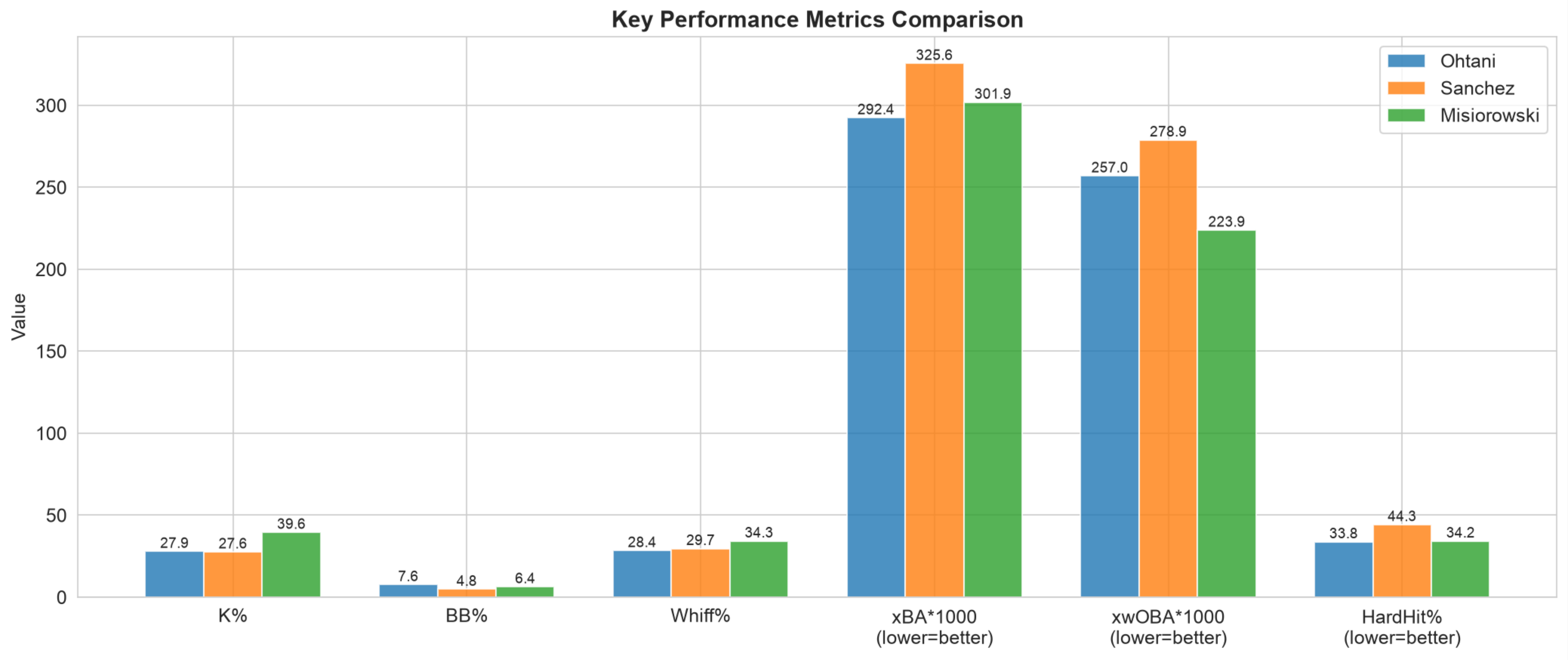
Sanchez is +14.5 runs above average.
Ohtani is +21.0 — he's cost his team more runs.

If Ohtani had Sanchez's RE rate over 1,335 pitches:
+10.7 total → would save ~10 additional runs.

Misiorowski's high total (+36.8) is surprising given his elite K%. This reflects high variance:
strikeout or walk/HR — little in between.

Key Metrics Comparison

Side-by-side look at the most important stats



Multi-axis comparison: K%, BB%, Whiff%, Chase%, Hard Hit%, GB rate, and RE/pitch.

Platoon Analysis: Sanchez Dominates LHB

An extreme reverse-split profile

Sanchez vs Left-Handed Batters

PA: 116

K%: 36.2% ← elite

BB%: 1.7% ← generational control

AVG: .138

OPS: .336 ← utterly dominant

HR: 1

Sanchez absolutely dominates LHB.
A 36.2% K% with only 1.7% BB% is
among the best platoon splits in baseball.

Sanchez vs Right-Handed Batters

PA: 380

K%: 25.0%

BB%: 5.8%

AVG: .263

OPS: .718

HR: 10

Sanchez is more vulnerable vs RHB (.718 OPS),
but his extreme ground ball rate limits damage.
10 HR in 380 PA is manageable for a sinker-baller.

Overall, Sanchez's platoon profile is viable
because LHB are virtually helpless against him.

Platoon Analysis: Misiorowski's Elite LHB K-Rate

47.6% K% vs lefties is extraordinary

Misiorowski vs Left-Handed Batters

PA: 229

K%: 47.6% ← generational

BB%: 7.4%

AVG: .087 ← almost unhittable

OPS: .288 ← elite, top-5 in MLB

HR: 2

Almost half of all LHB plate appearances end in a strikeout. This is elite.

Misiorowski vs Right-Handed Batters

PA: 193

K%: 30.1%

BB%: 5.2%

AVG: .192

OPS: .554

HR: 7

Still excellent vs RHB, but 7 HR in 193 PA is a concern. His fastball-heavy approach leaves him susceptible to pull-side power.

Overall: elite platoon profile with a slight vulnerability to right-handed power hitters.

Platoon Analysis: Ohtani's Relative Weakness

Less dominant vs both sides

Ohtani vs Left-Handed Batters

PA: 191

K%: 24.6%

BB%: 7.9%

AVG: .188

OPS: .545 ← good but not elite

HR: 3

vs LHB: .545 OPS is solid, but well behind Sanchez's .336 and Misiorowski's .288.

Ohtani's 24.6% K% vs LHB is notably low.

Ohtani vs Right-Handed Batters

PA: 149

K%: 32.2%

BB%: 7.4%

AVG: .128

OPS: .383 ← excellent vs RHB

HR: 1

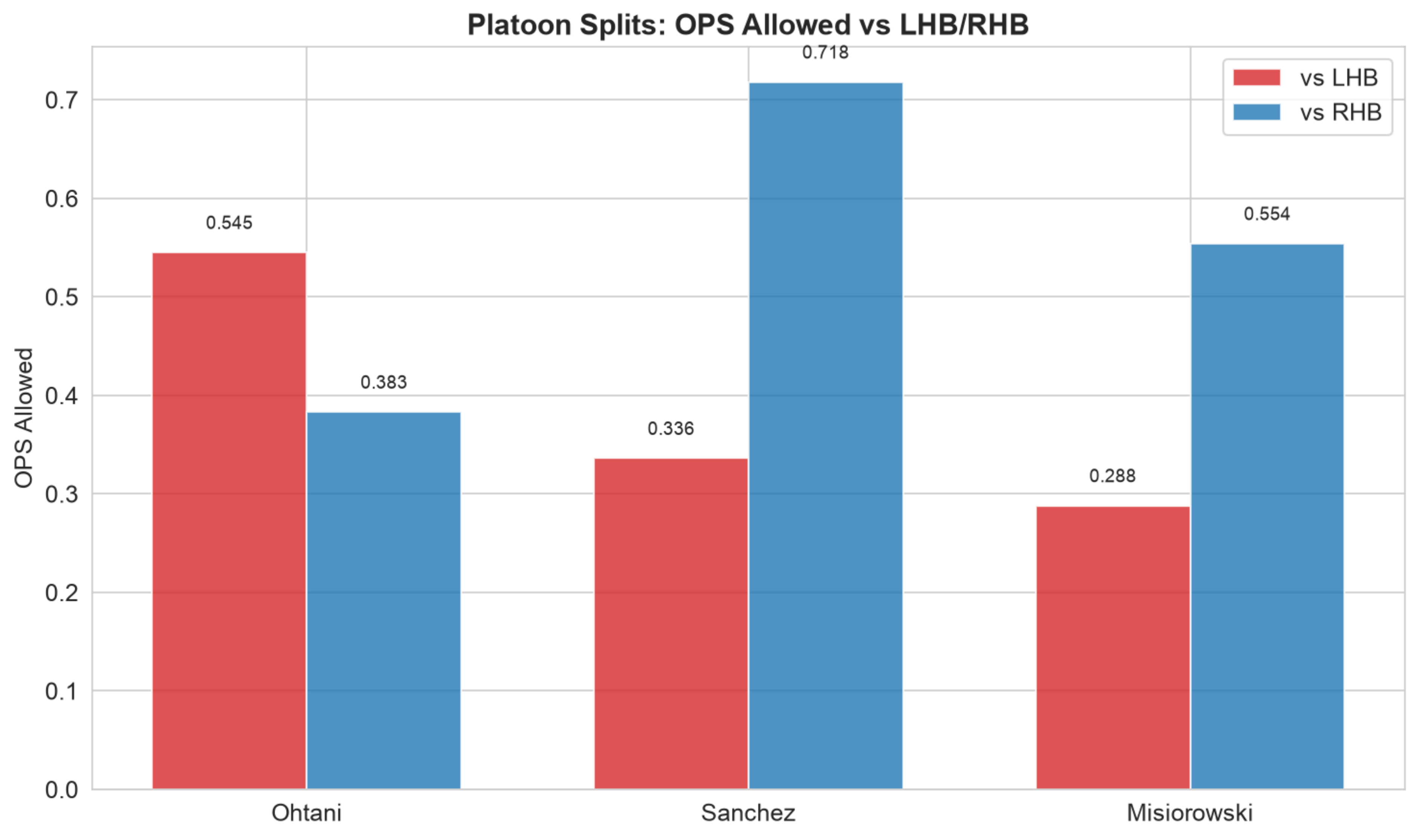
Ohtani handles same-handed batters well. .383 OPS with 32.2% K% is a strong split.

The gap: Ohtani vs LHB (.545 OPS) vs

Sanchez vs LHB (.336 OPS) = 62% worse.

Platoon Splits Visualization

OPS against by handedness



OPS allowed vs LHB (left) and vs RHB (right) for each pitcher. Lower is better.

Complete Platoon Comparison

Every split, every pitcher

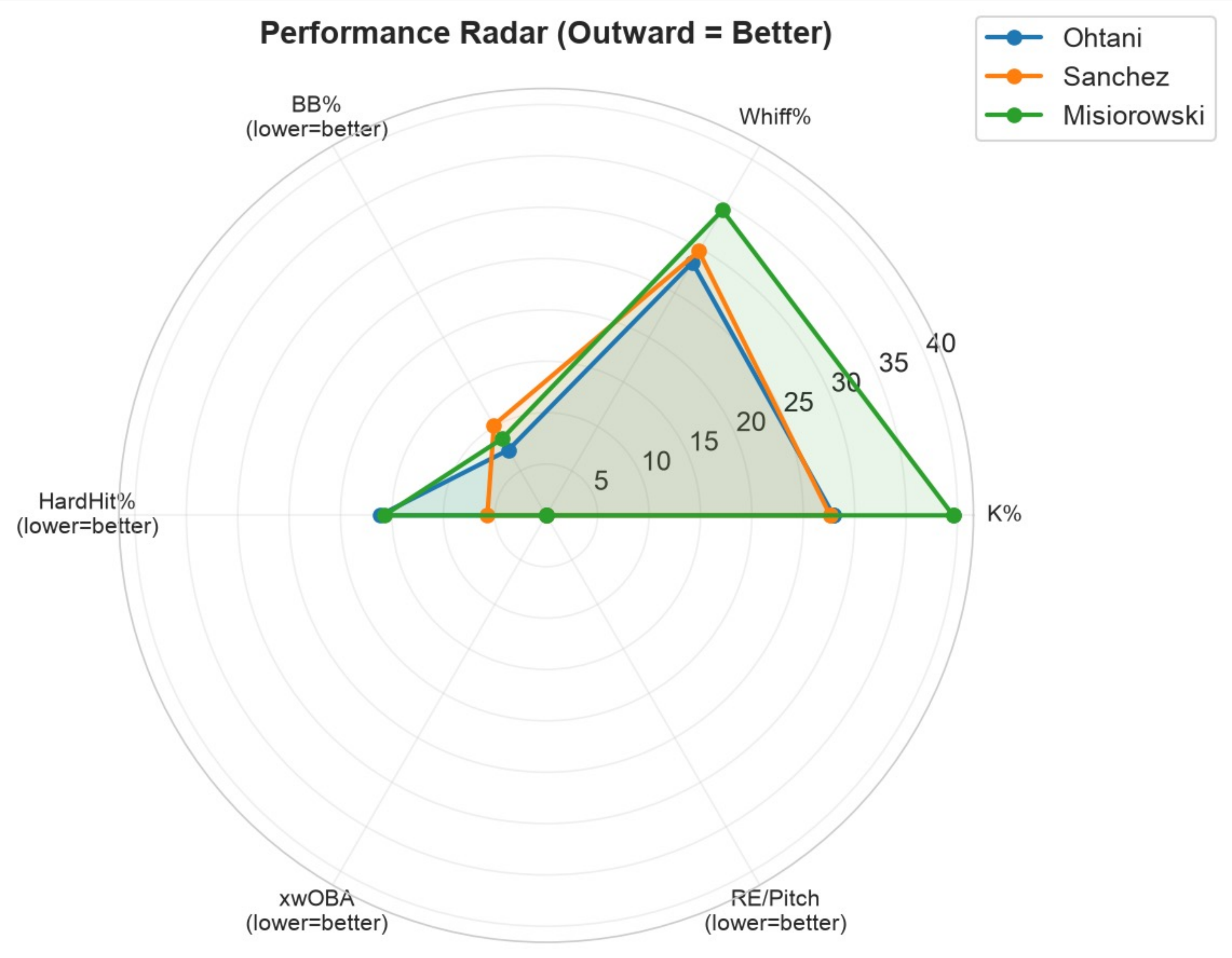
Pitcher	vs	PA	K%	BB%	AVG	OBP	SLG	OPS	HR
Ohtani	LHB	191	24.6%	7.9%	.188	.260	.285	.545	3
Ohtani	RHB	149	32.2%	7.4%	.128	.198	.185	.383	1
Sanchez	LHB	116	36.2%	1.7%	.138	.155	.181	.336	1
Sanchez	RHB	380	25.0%	5.8%	.263	.306	.412	.718	10
Misiorowski	LHB	229	47.6%	7.4%	.087	.162	.126	.288	2
Misiorowski	RHB	193	30.1%	5.2%	.192	.238	.316	.554	7

Summary:

- Sanchez dominates LHB (.336 OPS, 36.2% K%) — finest LHB-neutralizing in the group
- Misiorowski is unhittable vs LHB (.288 OPS, 47.6% K%) — generational platoon dominance
- Ohtani is strong vs RHB (.383 OPS) but mediocre vs LHB (.545) by this standard
- All three have exploitable platoon traits: Ohtani vs L, Sanchez vs R, Misiorowski vs R

Radar Comparison

Multi-dimensional performance profile



Radar chart comparing the three pitchers across 8 key dimensions. Larger area = more well-rounded performance.

Key Findings Summary

What the data tells us

Sanchez over Ohtani:

✓ 58% lower walk rate (4.8% vs 7.6%)
✓ 50% better run prevention per pitch (RE)
✓ Lower total run cost despite 465 more pitches
✓ Elite whiff rates on secondary offerings
✓ 192 vs 111 ground balls — damage suppression
✓ Far more durable (1,800 vs 1,335 pitches)

Misiorowski over both:

✓ 39.6% K% — clearly elite
✓ 34.3% Whiff% — best of the three
✓ 100.5 mph average fastball — 99th percentile
✓ 47.6% K% vs LHB — generational

Where Ohtani excels:

✓ 0 barrels allowed — genuine batted ball suppression
✓ 86.9 mph avg exit velocity — lowest of the 3
✓ Best expected stats (xwOBA .257)
✓ 7-pitch repertoire = adaptability

Where Ohtani falls short:

✗ Highest walk rate (7.6%) — command gap
✗ Highest run cost per pitch (+0.0158)
✗ Lowest overall whiff rate (28.4%)
✗ Lowest chase rate (29.8%) — less deception
✗ 35% fewer pitches — less total value
✗ Platoon vulnerability vs LHB (.545 OPS)

Conclusion

The data does NOT support the claim that Ohtani is the best pitcher of the three.

Sanchez > Ohtani in:

- Walk rate (4.8% vs 7.6%) — 58% better control
- Run prevention per pitch (+0.0080 vs +0.0158)
- Whiff rate on secondary pitches (42.7% vs 37.1%)
- Ground ball generation (192 GB vs 111 GB)
- Durability (1,800 vs 1,335 pitches)

Misiorowski > Both in:

- Strikeout rate (39.6%), Whiff rate (34.3%)
- Velocity (100.5 mph), LHB K% (47.6%)

Ohtani is excellent, but the data says he is the 3rd-best pitcher here.

Ohtani's strengths (0 barrels, low EV, good expected stats) do not translate to better actual run prevention.

Cy Young is about effectiveness and innings. By both measures, Sanchez and Misiorowski have stronger cases.

Data: MLB Statcast 2026 · Analysis: Python (pandas, numpy, matplotlib, seaborn, scipy)